

Fernando Simões

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Work Experience

Senior Platform Engineer

Nov 2025 - Present

Tintim

Remote

First platform engineer at a message attribution company (8M message events per day).

- Built and operate a stateful WebSocket platform on self-hosted Kubernetes using StatefulSets, with capacity-aware scheduling (sessions per pod, pods per node).
- Wrote a Go gateway that routes traffic by session: maintains an in-memory session-to-pod map converged via periodic reconciliation against the cluster, places new sessions on the least-loaded pod, and enforces per-pod capacity limits; runs as stateless replicas with no shared store.
- Traced a recurring Postgres outage to query patterns and I/O saturation, and characterized autovacuum and Postgres performance under real workloads (frn.sh/pgfsync).
- Stabilized pod memory by profiling `/proc/pid/smmaps_rollup`, identifying V8 heap and glibc arena fragmentation, increasing pod density per node and halving infrastructure cost (frn.sh/pmem); also diagnosed Python GIL contention in production workloads.
- Built a two-layer observability setup separating operational monitoring (Prometheus, Grafana, Alertmanager, Loki) from an analytical pipeline (Firehose, Athena, Glue).
- Introduced Infrastructure as Code and migrated existing services to Terraform.

Senior Platform Engineer

Jun 2025 - Nov 2025

GCB Investimentos

Remote

Senior Platform Engineer on a 4-person central team serving product, data, and AI teams across a 250-person fintech.

- Shipped a Go CLI adopted across engineering for local development against production-like environments (Docker Compose, LocalStack, sandboxed RDS snapshots).
- Designed and rolled out company-wide AWS WAF rules with centralized rule management and automated threat escalation, surfacing previously undetected anomalous traffic patterns.
- Diagnosed and fixed cross-service production issues including socket leaks from unconsumed S3 response bodies and Redis memory exhaustion.

Software Engineer

Feb 2024 - Jun 2025

Liber

Remote

Built greenfield fintech platform for distributed receivables assignment, with projected R\$1M monthly revenue.

- Developed microservices using Go and Postgres on AWS, implementing business logic for receivables processing and financial data validation.
- Built infrastructure components using Terraform and AWS, setting up observability with Datadog across microservices handling financial transactions.

Software Engineer

Mar 2021 - Feb 2024

Cingulo

Remote

Backend development for a mental health SaaS platform. Python, Django REST Framework, Postgres, Celery, and AWS.

- Led development of the payment gateway, enabling subscription payments outside of mobile app stores, increasing conversion and reducing dependency on in-app purchases.
- Implemented CI/CD with GitHub Actions, standardized testing with Pytest, and migrated internal tools to a modern stack.

Skills

Languages: Go, Python, Bash

Infrastructure: AWS, Terraform, Terragrunt, Kubernetes, FluxCD, Docker

Observability: Prometheus, Grafana, Loki, Datadog, eBPF, perf, flame graphs

Databases: PostgreSQL, Redis

Other: Git, GitHub Actions, Azure DevOps, Linux internals, TCP/IP and computer networks